HW05 Performance Test Report

Student ID 23127326 | Local run date 2026-08-30

1. Execution context

Field	Value
SUT	http://localhost:3000 | commit 85af3ba875c88283615e22cb108f13e2fccaf0e9
Host	MacBook Pro 18,3 | Apple M1 Pro | 10 cores | 32 GB RAM | macOS 26.5.2
Tool	Apache JMeter 5.6.3 | Java 21 | headless non-GUI run

2. Workflow

Lockout probe -> valid login -> product search -> cart add -> cart quantity update -> cart read -> checkout -> post-checkout cart assertion. Each official VU uses a distinct synthetic account and a per-VU CSV file.

3. Scenario results

Scenario	Samples	HTTP err	Assertion gap	p95 overall / max label	TPS
Load	3,287	0 (0.00%)	356 (10.83%)	6 / 7 ms	9.1583
Stress	16,433	0 (0.00%)	1,780 (10.83%)	5 / 7 ms	34.4137
Spike	7,171	0 (0.00%)	751 (10.47%)	5 / 7 ms	17.1228
Endurance	24,574	0 (0.00%)	2,699 (10.98%)	6 / 8 ms	30.4315

Weighted mean: Load 2.642 ms; Stress 1.914 ms; Spike 2.041 ms; Endurance 2.443 ms. Resource monitor: CPU max 16.4-25.5%; RSS max 75.8-120.8 MB; 11 threads.

4. Interpretation

All failed samples are the explicit POST_CHECKOUT_CART - expected empty assertion. The SUT leaves the cart non-empty after checkout. Therefore the 10.47-10.98% failure rates are business-gap evidence, not transport or HTTP failure. Response times are below the provisional 1,000 ms p95 threshold.

Resource rerun used the corrected macOS monitor. Endurance RSS ranged 70.1-85.6 MB in this 13.5-minute local run; no leak conclusion is claimed beyond this sample.

5. Reproduced SUT gaps

Gap	Evidence
Lockout	Four responses: 401, 401, 403, 403; DB attempts 4; locked window 180 s. See evidence/issues/lockout-probe-20260830.jsonl.
Checkout cleanup	Post-checkout assertion fails for every observed checkout in all four official workloads.
Pagination / duplicate line	Implementation candidates retained; independent response probe still required before external issue.

6. Reproduction and evidence

Raw JTL and JMeter HTML are under results/{load,stress,spike,endurance}. Plans are under test-plans. Synthetic data and provisioning scripts are under data/ and tools/. AI audit, critique and continuous-testing proposal are included. Video and GUI screenshots remain manual tasks for the student.